

SINGLE-CELL TRANSCRIPTOMICS

End-to-End Breast Cancer scRNA-seq Portfolio Project

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Executive Summary

This portfolio project implements an end-to-end single-cell RNA-sequencing (scRNA-seq) workflow in Python/Scanpy using a public breast-cancer expression dataset. The analysis begins with a 21,785-gene by 1,534-cell expression matrix, performs quality control, normalization, highly-variable-gene selection, PCA, neighborhood graph construction, UMAP visualization, Leiden clustering, marker-gene analysis, attempted cell-type annotation, differential-expression analysis, and biological signature scoring.

Most important finding from the current run

The computational clustering is successful and reveals 16 transcriptionally distinct clusters. Marker expression clearly supports several biological identities (epithelial-like, T-cell, B-cell, NK, myeloid/macrophage, fibroblast/stromal and endothelial-like regions). However, the automated/manual cell_type field was never populated beyond 'Unknown'. Therefore, the cell-type proportion plot and the 'Unknown vs. rest' differential-expression plot are not biologically interpretable yet. The correct next step is to map cluster IDs to validated cell-type labels and rerun those two analyses.

Accordingly, this report separates valid results from provisional results. It does not treat the all-'Unknown' annotation as a biological conclusion.

Project Objectives

- Build a reproducible Python-based scRNA-seq workflow suitable for a bioinformatics portfolio.
- Identify transcriptionally distinct cell populations in breast-cancer single-cell data.
- Use canonical marker genes to infer major epithelial, immune, myeloid, stromal and endothelial populations.
- Assess patient/sample contributions to the UMAP structure.
- Explore tumor-microenvironment programs including T-cell exhaustion, Treg activity, pro-tumor macrophage signaling, TNF/NF- κ B inflammation, fibroblast extracellular-matrix activity and hypoxia.
- Create publication-style figures and identify limitations that must be resolved before making condition-level biological claims.

1. Analysis Workflow

Stage	Method	Purpose
Data preparation	Pandas + AnnData	Load expression matrix and organize cell/gene metadata.
Quality control	Scanpy QC metrics	Detect low-complexity cells and technical outliers.
Normalization	Library-size normalization + log1p	Make expression values more comparable across cells.
Feature selection	Highly variable genes	Focus dimensionality reduction on informative genes.
Dimensionality reduction	PCA	Compress major expression variation.
Cell neighborhood	kNN graph	Model transcriptional similarity between cells.

Visualization	UMAP	Represent cell-state structure in two dimensions.
Clustering	Leiden	Identify transcriptionally related groups.
Annotation	Canonical markers + dot plot	Assign biological meaning to clusters.
DEG analysis	Wilcoxon rank-sum	Find genes enriched in each cluster/group.
Functional analysis	Gene signatures / enrichment	Interpret biological programs.

2. Quality Control

Quality control evaluates whether each cell contains enough informative RNA and whether technical artifacts may dominate the dataset. Three metrics were plotted: number of detected genes per cell, total counts per cell, and mitochondrial-count percentage.

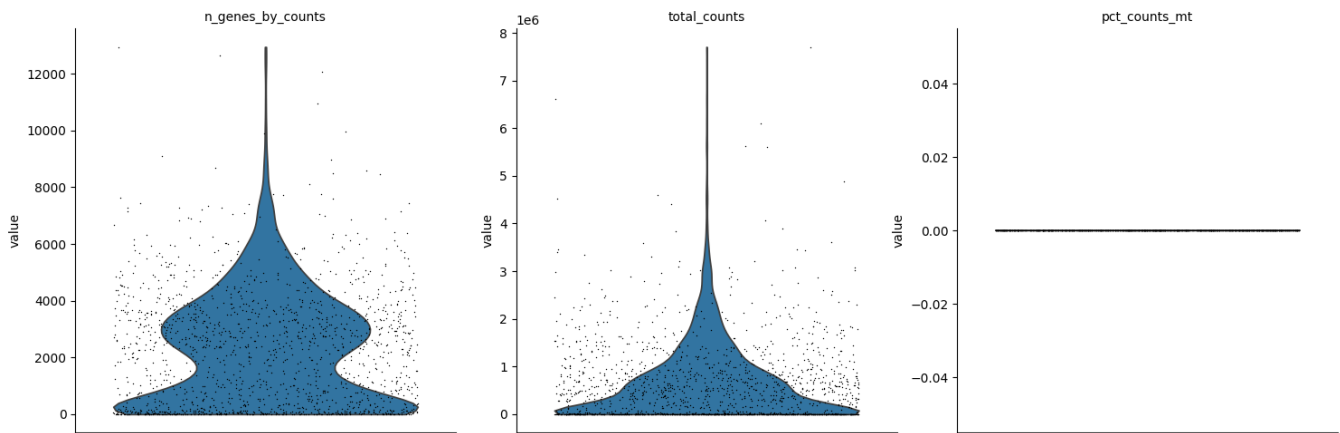


Figure 1. QC distributions for detected genes, total counts and mitochondrial percentage.

Interpretation

- `n_genes_by_counts` has a broad distribution. Many cells fall in the low-complexity range, while another major population contains roughly a few thousand detected genes. A small number of cells extend above 8,000-12,000 genes and may represent unusually complex cells or potential multiplets.
- `total_counts` is strongly right-skewed: most cells have lower counts, while a small number have very high totals. This is common in single-cell data and motivates normalization.
- `pct_counts_mt` is exactly 0 across the plot. This does not prove that every cell has zero mitochondrial RNA. More likely, mitochondrial genes were not detected by the 'MT-' naming rule in this processed RSEM matrix or mitochondrial features are absent/represented differently. Therefore, mitochondrial filtering was not informative in this run.

QC caution

Because mitochondrial percentage is uniformly zero, do not state that the dataset contains no stressed cells. Instead report that mitochondrial QC could not be evaluated reliably from the available gene naming/processed matrix.

3. Principal Component Analysis (PCA)

PCA compresses thousands of gene-expression measurements into principal components (PCs). PC1 captures the largest direction of variation, followed by PC2, PC3 and so on. The plotted values are shown on a logarithmic scale, which is why the y-axis contains negative values.

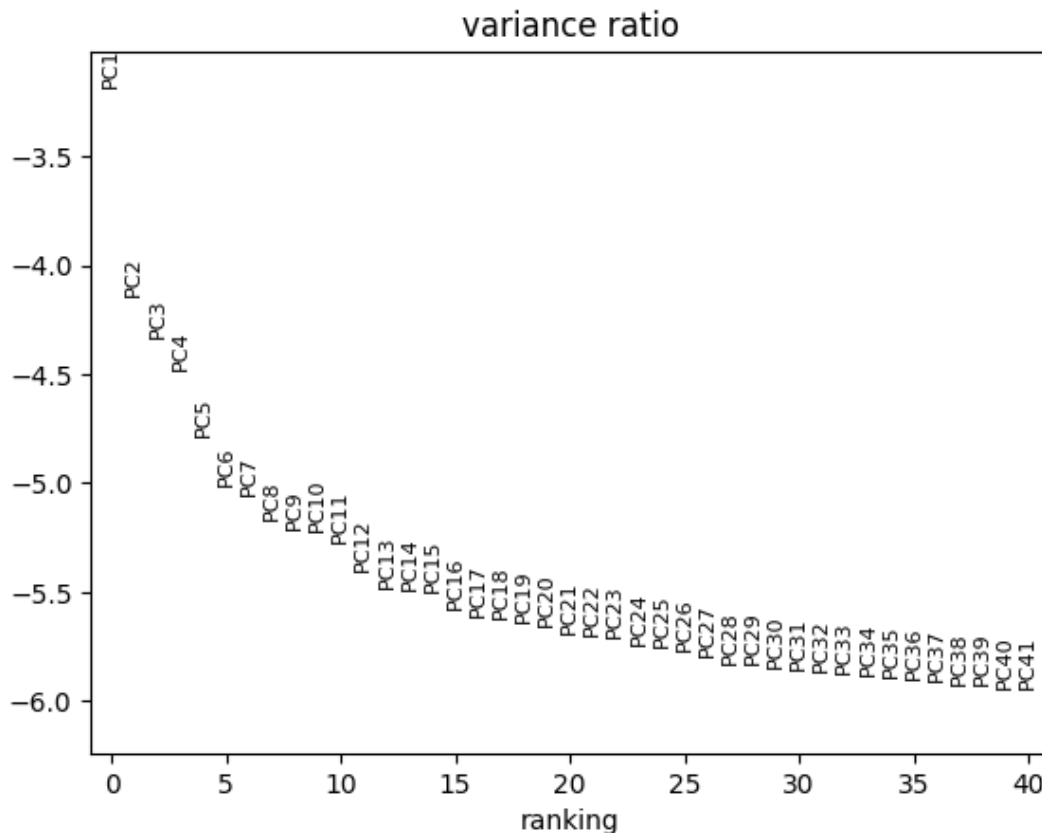


Figure 2. PCA variance-ratio ranking across the first principal components.

The curve falls sharply across the earliest PCs and then gradually flattens. This indicates that the first components contain the strongest structure, while later PCs contribute progressively smaller amounts of additional variation. Using approximately the first 30 PCs for neighbor construction is therefore a reasonable exploratory choice, although a formal elbow choice could be refined.

4. UMAP and Leiden Clustering

UMAP places transcriptionally similar cells close together and dissimilar cells farther apart. Leiden clustering then partitions the neighborhood graph into discrete groups. In this analysis, clusters 0-15 were identified, giving 16 computational populations.

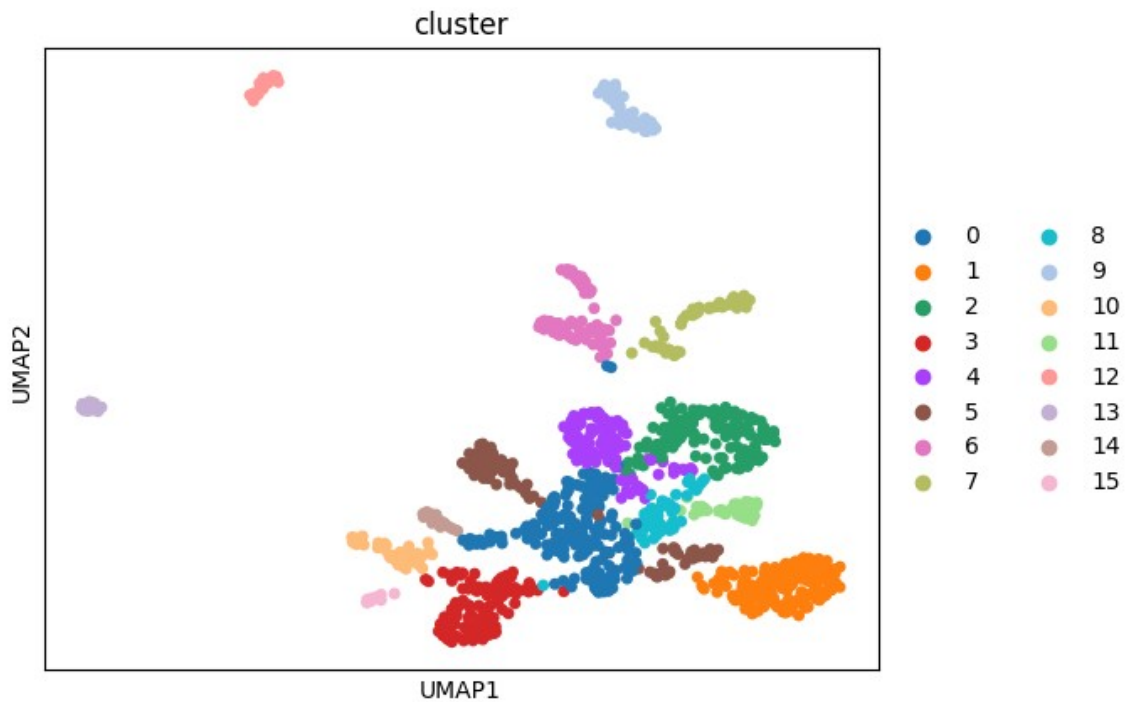


Figure 3. UMAP colored by Leiden cluster. Sixteen transcriptional clusters are visible.

What the graph shows

- Several clusters form compact, well-separated islands, suggesting strongly distinct transcriptional programs.
- Other clusters lie close together in the central/lower UMAP region, suggesting related cell states or gradual transitions.
- A cluster is not automatically a cell type. Cluster identity must be supported by marker genes.

5. Patient and Sample-Group Distribution

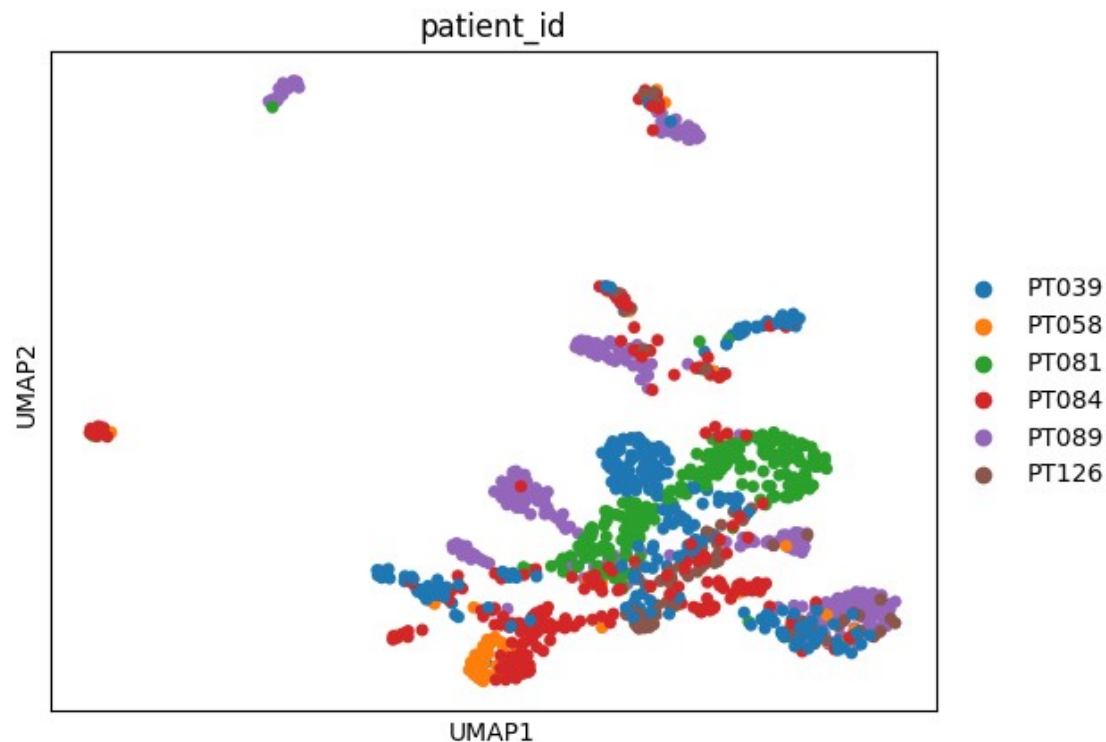


Figure 4. UMAP colored by patient ID.

Cells from PT039, PT084 and PT089 occupy multiple regions, whereas some compact islands are enriched for particular patients. This means the UMAP contains both shared biological structure and patient-specific structure. A few regions appear strongly patient-associated, so patient/batch effects should be considered before condition-level comparisons.

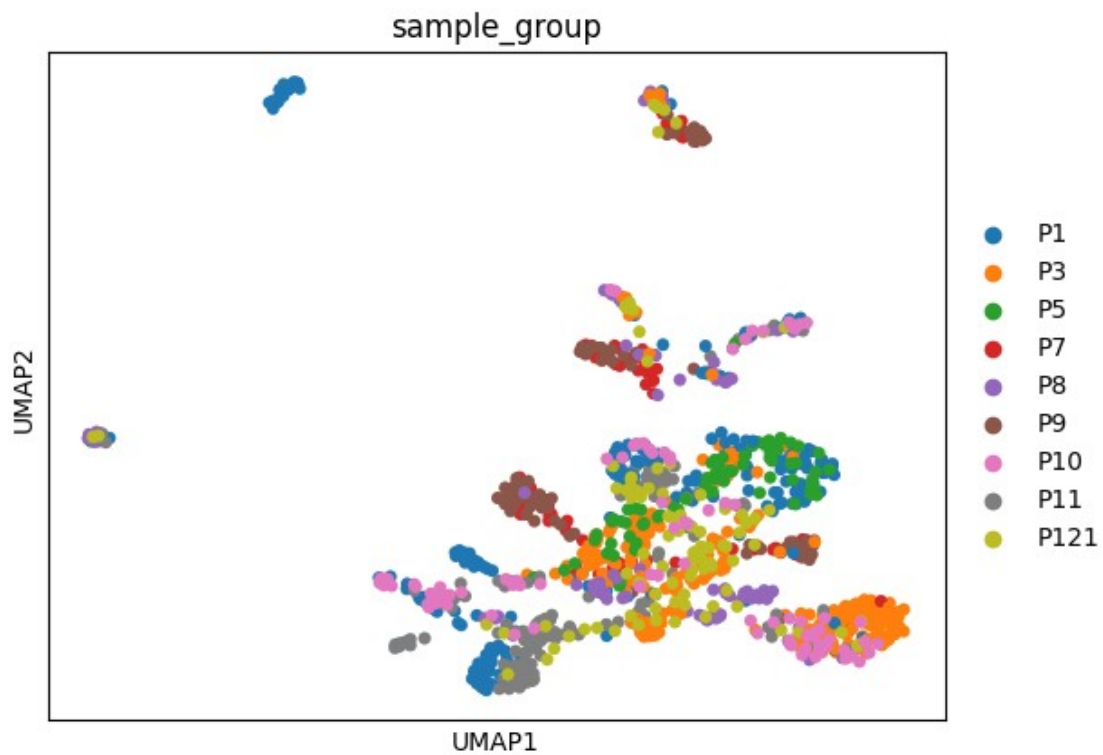


Figure 5. UMAP colored by sample group.

Sample groups are mixed across much of the central UMAP, which is encouraging, but several peripheral islands are enriched for particular sample groups. This pattern suggests that sample identity contributes to some clusters and should be accounted for in a more advanced analysis using integration/batch-correction methods where appropriate.

6. Canonical Marker-Gene Expression

Canonical markers were projected onto the UMAP to determine which regions resemble known cell populations. Bright/yellow regions indicate higher expression; dark/purple regions indicate lower expression.

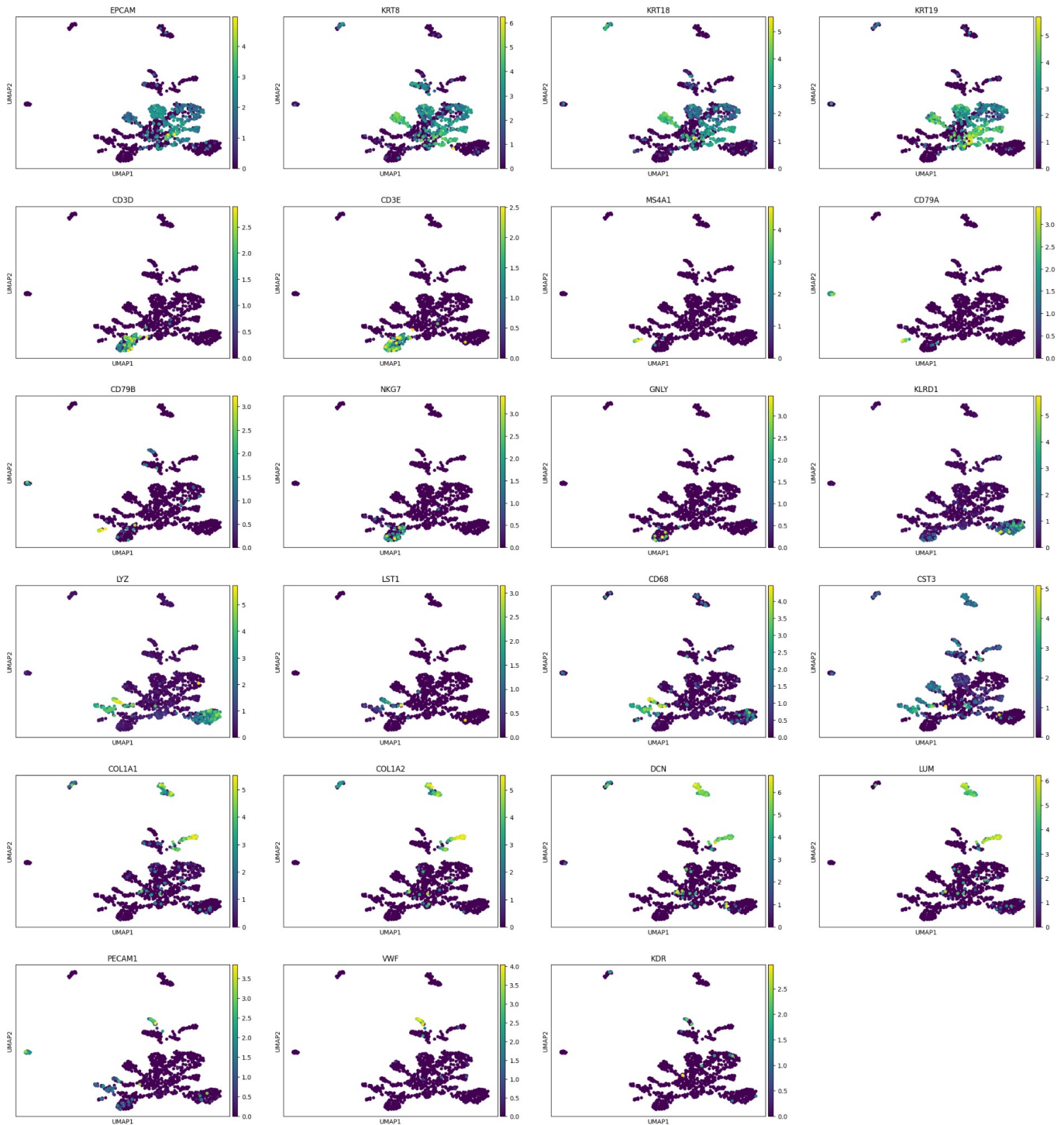


Figure 6. UMAP feature plots for canonical epithelial, immune, myeloid, stromal and endothelial markers.

Biological interpretation of the marker panels

- Epithelial/malignant-like markers: EPCAM, KRT8, KRT18 and KRT19 are concentrated in subsets of the lower/central UMAP, supporting epithelial-like populations.
- T-cell markers: CD3D and CD3E strongly localize to a compact lower-left region, providing clear evidence for a T-cell cluster.
- B-cell markers: MS4A1, CD79A and CD79B are concentrated in a distinct small region, supporting a B-cell population.

- NK-cell markers: NKG7, GNLY and KLRD1 show a cytotoxic lymphoid pattern; KLRD1 is especially strong in a right-lower island.
- Myeloid/macrophage markers: LYZ, LST1, CD68 and CST3 are prominent in multiple lower/right regions, indicating heterogeneous myeloid populations.
- Fibroblast/stromal markers: COL1A1, COL1A2, DCN and LUM strongly mark upper/peripheral islands and parts of the central structure, supporting stromal/fibroblast populations.
- Endothelial markers: PECAM1, VWF and KDR mark a smaller subset, supporting an endothelial-like population.

7. Marker Dot Plot and Cluster-Level Annotation

The dot plot summarizes the same marker information by cluster. Dot size represents the fraction of cells in a cluster expressing the gene, while color intensity represents mean expression. This is one of the most useful figures for assigning cluster identities.

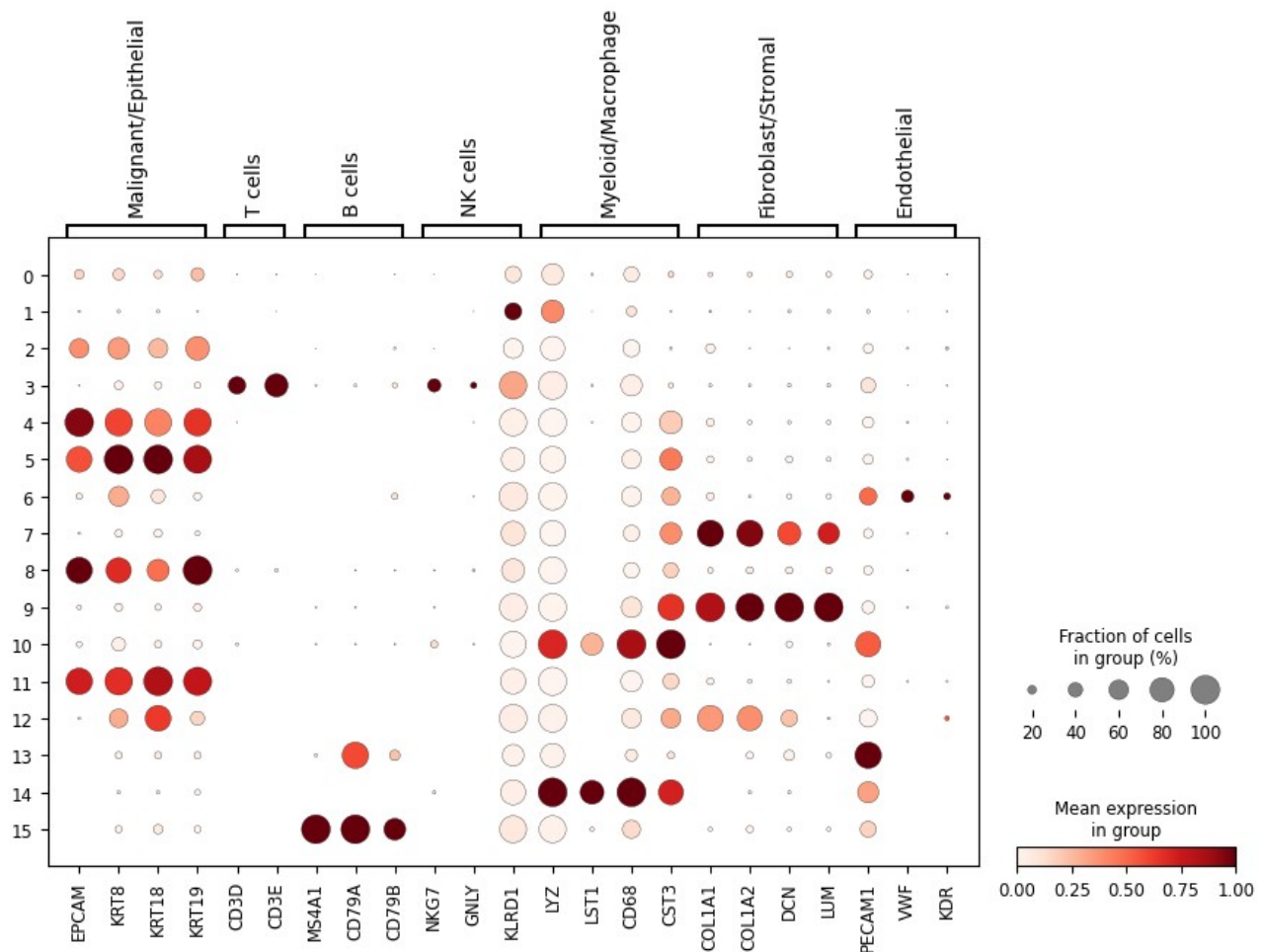


Figure 7. Canonical marker dot plot across clusters 0-15.

Strong signals visible in the dot plot

- Clusters 4, 5, 8 and 11 show strong keratin/epithelial-marker expression, supporting epithelial-like identities.

- Cluster 3 shows strong CD3D/CD3E with NKG7 signal, supporting a T/cytotoxic lymphoid identity.
- Cluster 15 shows strong MS4A1/CD79A/CD79B, strongly supporting a B-cell identity.
- Clusters 1, 10 and 14 show strong LYZ/LST1/CD68-related myeloid signal, supporting myeloid/macrophage-like identities.
- Clusters 7 and 9 show strong COL1A1/COL1A2/DCN/LUM, strongly supporting fibroblast/stromal identities.
- Cluster 13 shows strong PECAM1 with endothelial-marker signal, supporting an endothelial identity.
- Some clusters show mixed markers and should remain provisional until a more systematic annotation is performed.

8. Differential Marker Genes by Cluster

Wilcoxon ranking identified genes that distinguish each cluster from all remaining cells. These genes provide an independent way to validate the canonical-marker interpretation.

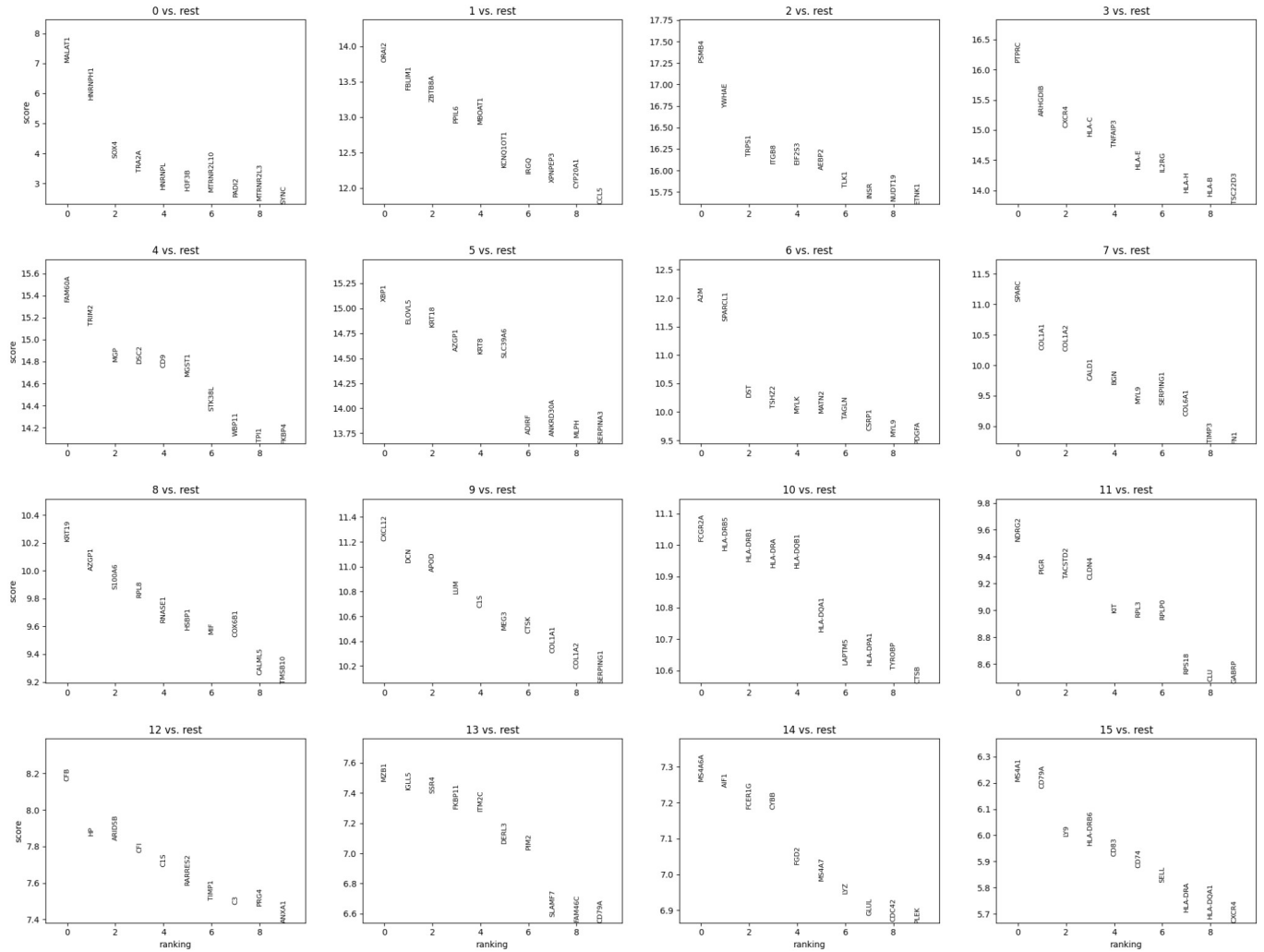


Figure 8. Top ranked genes for clusters 0-15 versus the rest of the dataset.

Examples from the ranked genes

- Cluster 3 includes CD3D, CD3E and NKG7 among its top markers, reinforcing a T/NK-like immune interpretation.
- Cluster 7 contains COL1A1 and COL1A2, consistent with fibroblast/extracellular-matrix biology.
- Cluster 8 contains KRT19 and AZGP1, consistent with epithelial characteristics.
- Cluster 10 includes FCGR2A, HLA-DRA/HLA-DRB1 and other antigen-presentation genes, consistent with an immune/myeloid population.
- Cluster 15 includes MS4A1, CD79A, CD83 and CD74, supporting a B-cell/antigen-presenting identity.
- Cluster 14 includes LYZ and FCER1G, consistent with myeloid lineage.

These cluster-level results are biologically informative even though the later `cell_type` field remained 'Unknown'. They should be used to build the final cluster-to-cell-type mapping.

9. Current Cell-Type Annotation: Critical Issue

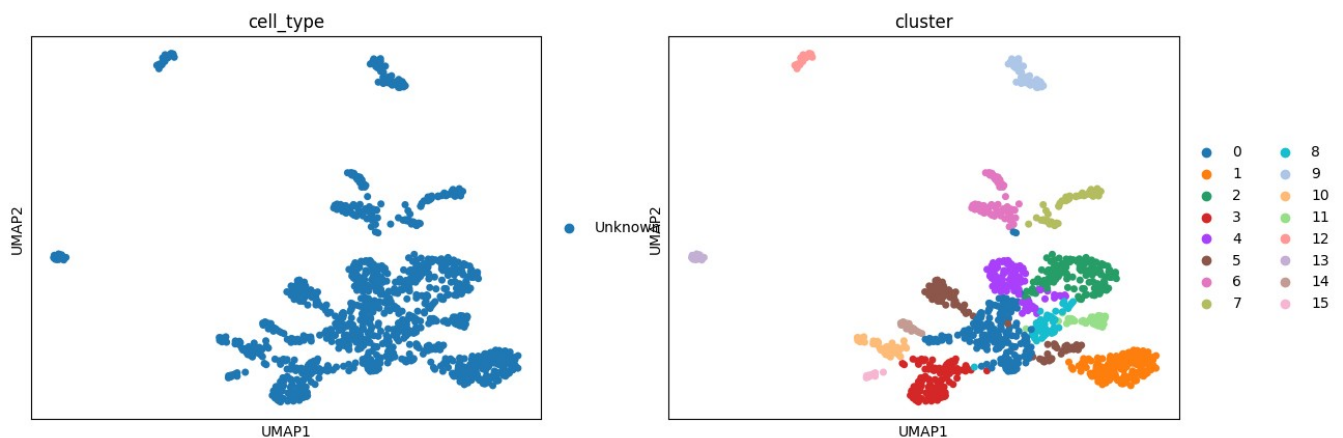


Figure 9. Current `cell_type` UMAP (left) versus valid Leiden clusters (right).

The left panel labels every cell as 'Unknown'. This is not evidence that the cells are biologically unidentifiable. It means the cluster-to-cell-type dictionary in the annotation script was not updated with the identities inferred from the marker plots.

Required correction before final biological reporting

Assign validated labels to cluster IDs (for example, T-cell, B-cell, myeloid/macrophage, fibroblast/stromal, endothelial and epithelial-like groups) based on the dot plot and ranked markers. Then save a corrected AnnData object and rerun cell-type proportions, cell-type DEG and any cell-type-specific pathway analysis.

10. Cell-Type DEG and Proportion Outputs: Why They Are Not Yet Valid

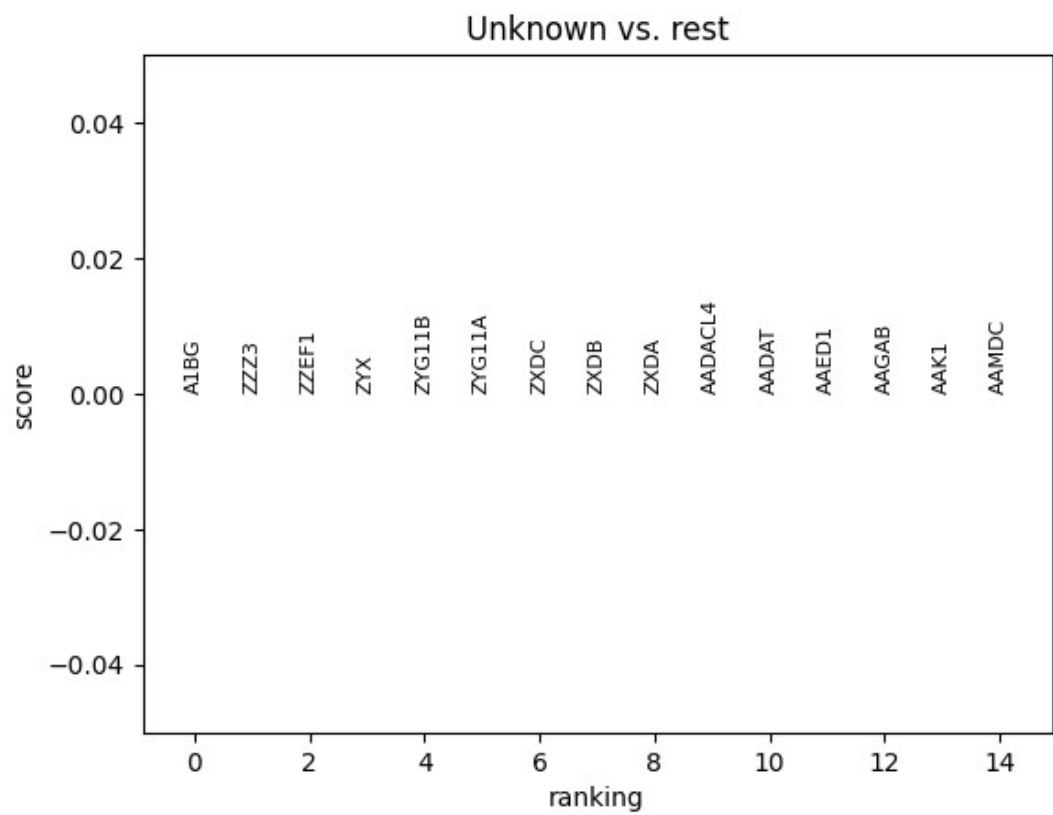


Figure 10. 'Unknown vs. rest' DEG output. The near-zero scores reflect the failed one-category annotation.

Because all cells share the same cell_type label ('Unknown'), there is no meaningful second biological group for comparison. The resulting 'Unknown vs. rest' plot therefore contains near-zero scores and must not be interpreted as a biological DEG result.

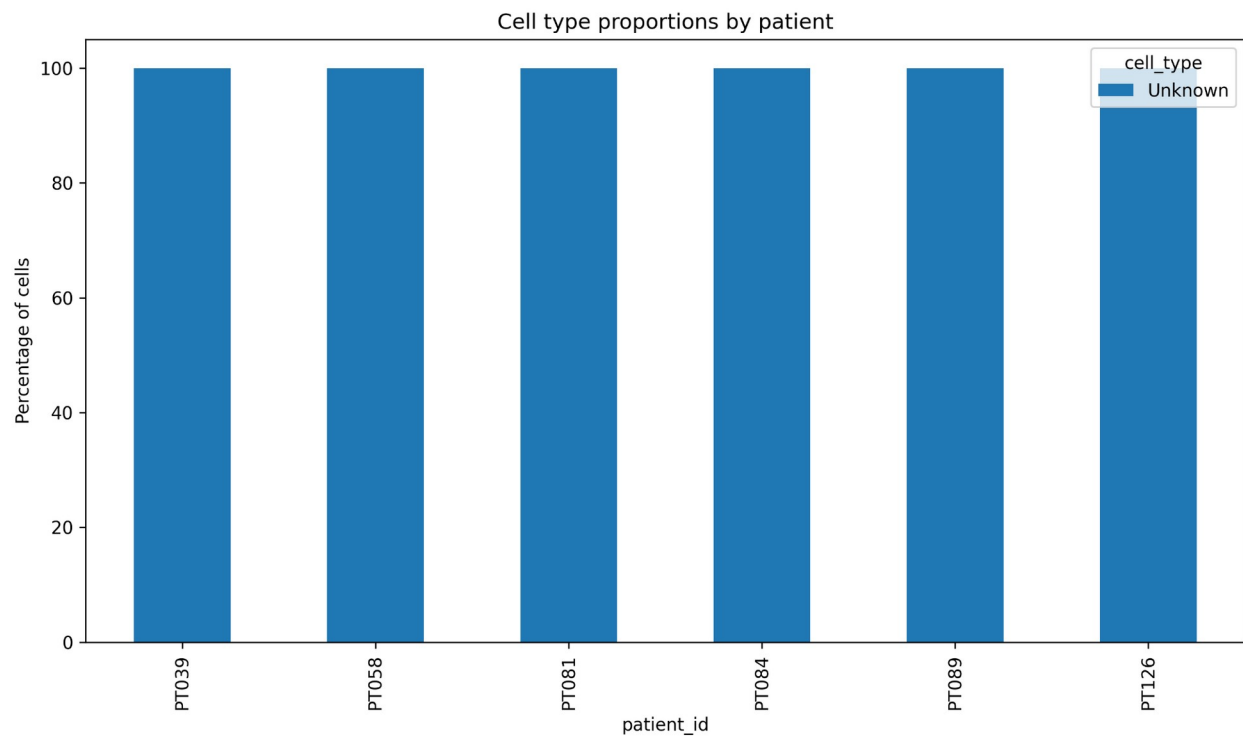


Figure 11. Cell-type proportions by patient. Every bar is 100% Unknown because annotation was not assigned.

Likewise, every patient appears as 100% Unknown. This plot is technically correct for the current metadata but biologically uninformative. After cluster annotation, this should become a stacked composition plot showing the relative proportions of epithelial, immune, myeloid, stromal and endothelial populations in each patient.

11. Biological Signature Scoring

Gene-set scoring was used to explore six programs inspired by tumor-microenvironment biology: T-cell exhaustion, regulatory T-cell activity, pro-tumor macrophage activity, TNF/NF- κ B inflammation, fibroblast extracellular-matrix activity and hypoxia. Scores are relative within this dataset: brighter regions have higher expression of the genes used in that signature.

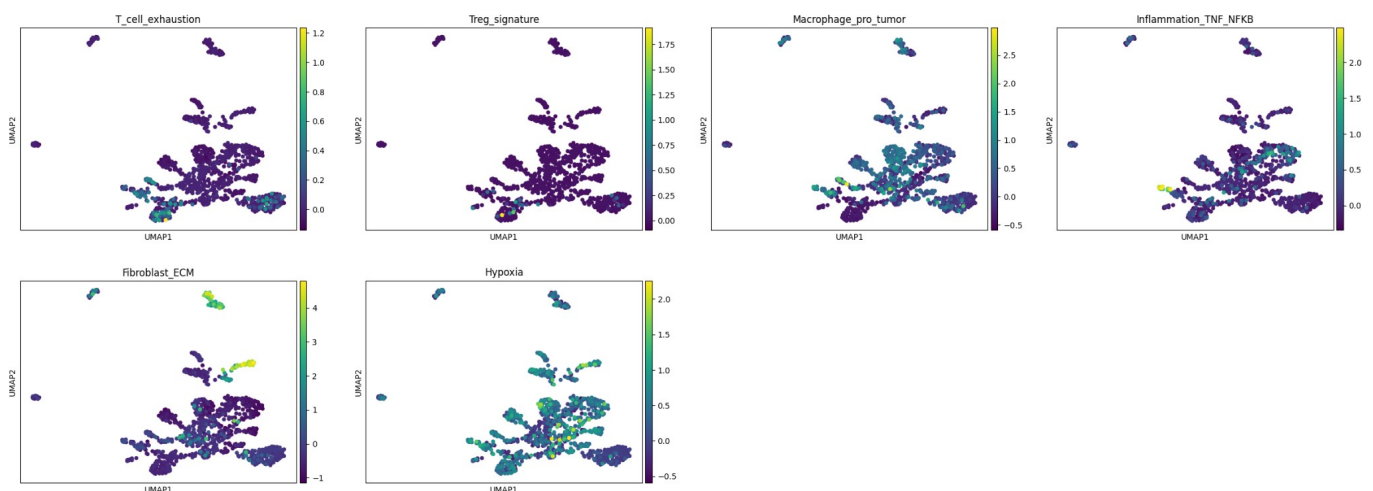


Figure 12. UMAP signature scores for immune, stromal, inflammatory and hypoxia-related programs.

Interpretation of each signature

- T-cell exhaustion: elevated mainly in a lower-left immune region, consistent with the location of CD3D/CD3E-positive cells. This suggests a subset of T cells expresses exhaustion-associated genes.
- Treg signature: strongly concentrated in a small lower-left subset, suggesting a focused regulatory-T-cell-like program rather than a dataset-wide signal.
- Macrophage pro-tumor: broadly elevated across several central/lower regions with local hotspots, consistent with heterogeneous myeloid/macrophage activity.
- Inflammation TNF/NF- κ B: shows localized hotspots rather than uniform activation, suggesting inflammatory signaling is concentrated in specific cellular states.
- Fibroblast ECM: strongest in upper/peripheral islands that also express COL1A1/COL1A2/DCN/LUM, strongly supporting stromal extracellular-matrix biology.
- Hypoxia: distributed across several central and peripheral populations, indicating that hypoxia-associated transcription is not restricted to a single cluster.

These signature scores are exploratory. They support biological interpretation but do not replace formal cell-type annotation or statistical comparison between clinically defined conditions.

12. Integrated Biological Interpretation

Taken together, the valid outputs demonstrate substantial cellular heterogeneity in this breast-cancer single-cell dataset. The 16 Leiden clusters are supported by coherent marker-gene patterns. Distinct epithelial-like, T-cell, B-cell, cytotoxic/NK-like, myeloid/macrophage, fibroblast/stromal and endothelial-like compartments can be recognized from the canonical markers and cluster-ranked genes.

The tumor microenvironment is therefore not a single homogeneous population. Instead, it contains multiple immune and stromal states alongside epithelial-like cells. The signature maps further suggest spatially distinct transcriptional programs related to T-cell dysfunction/regulation, macrophage biology, inflammation, extracellular-matrix remodeling and hypoxia.

What this project successfully demonstrates

You successfully built and executed the computational backbone of a real scRNA-seq analysis: data ingestion, QC, normalization, dimensionality reduction, graph construction, clustering, marker analysis and functional scoring. The main unfinished scientific step is converting cluster numbers into validated cell-type labels and then rerunning downstream cell-type comparisons.

13. Relationship to the Reference Study

This portfolio project is an analytical mimic rather than an exact reproduction of the reference ER-positive primary-versus-metastatic breast-cancer study. The reference study used a larger, specifically designed cohort and performed additional analyses such as malignant-cell CNV inference, finer immune/stromal subclustering, regulatory-network analysis and cell-cell communication. Your project reproduces the core

computational logic of identifying cell states from single-cell expression and interpreting tumor-microenvironment programs.

14. Limitations

- The current cell_type annotation is incomplete; all cells are stored as Unknown.
- Therefore the existing patient cell-type proportion plot and cell-type DEG plot are not valid biological summaries.
- Mitochondrial QC is non-informative because pct_counts_mt is zero across cells under the current MT-gene-name rule.
- Some UMAP regions are enriched for particular patients/sample groups, so batch/patient effects may contribute to clustering.
- Canonical marker annotation alone may be insufficient for ambiguous or mixed clusters; reference-based annotation or additional marker panels would improve confidence.
- Signature scores are exploratory and depend on the selected genes; they should not be interpreted as proof of pathway activation.
- This dataset/workflow should not be described as an exact reproduction of the reference paper's primary-versus-metastatic comparison.

15. Recommended Next Steps

Step	Action	Purpose
1	Finalize cluster annotation	Create a cluster_to_celltype mapping using the dot plot and top-ranked genes. Keep ambiguous clusters as 'Unresolved' rather than forcing a label.
2	Rerun composition analysis	Generate corrected cell-type proportions by patient and sample.
3	Rerun DEG analysis	Compare biologically valid cell types or clinically defined groups only after labels/metadata are correct.
4	Add pathway enrichment	Use cluster/cell-type marker genes with GO, KEGG or Hallmark sets and report adjusted P-values.
5	Consider batch integration	Evaluate Harmony, BBKNN or scVI if patient/sample effects are strong and integration is scientifically appropriate.
6	Advanced paper-mimic modules	Add malignant-cell CNV inference, immune/stromal subclustering and ligand-receptor communication as

		optional extensions.
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16. Portfolio Summary

This project demonstrates practical experience with Python, Pandas, AnnData, Scanpy, dimensionality reduction, graph-based clustering, single-cell visualization, marker-gene interpretation and biological gene-set scoring. It also demonstrates an important research skill: recognizing when an output is computationally generated but not biologically valid. Rather than presenting the all-'Unknown' cell-type results as findings, the analysis identifies annotation as the next correction required.

Suggested Portfolio Description

Developed an end-to-end single-cell RNA-seq analysis workflow in Python/Scanpy for a public breast-cancer dataset, including QC, normalization, HVG selection, PCA, UMAP, Leiden clustering, canonical marker analysis, cluster-level differential expression and tumor-microenvironment signature scoring. Identified 16 transcriptional clusters and resolved major epithelial, lymphoid, myeloid, stromal and endothelial marker patterns, while auditing annotation and batch-related limitations before downstream interpretation.

Appendix: Figure Guide

Figure	Output	Meaning
Figure 1	QC violin plots	Cell complexity, library size and mitochondrial metric.
Figure 2	PCA variance ratio	Information captured by successive PCs.
Figure 3	Leiden UMAP	Sixteen transcriptional clusters.
Figure 4	Patient UMAP	Patient contribution to UMAP structure.
Figure 5	Sample-group UMAP	Sample-group mixing/separation.
Figure 6	Marker feature plots	Spatial expression of canonical cell markers.
Figure 7	Marker dot plot	Cluster-level marker prevalence and mean expression.
Figure 8	Cluster marker ranking	Top genes distinguishing each cluster.
Figure 9	Cell-type annotation UMAP	Shows annotation has not yet been assigned.
Figure 10	Cell-type DEG	Invalid/provisional because all labels are Unknown.
Figure 11	Patient proportions	Invalid/provisional because all labels are Unknown.
Figure 12	Signature scores	Exploratory immune/stromal/hypoxia programs.